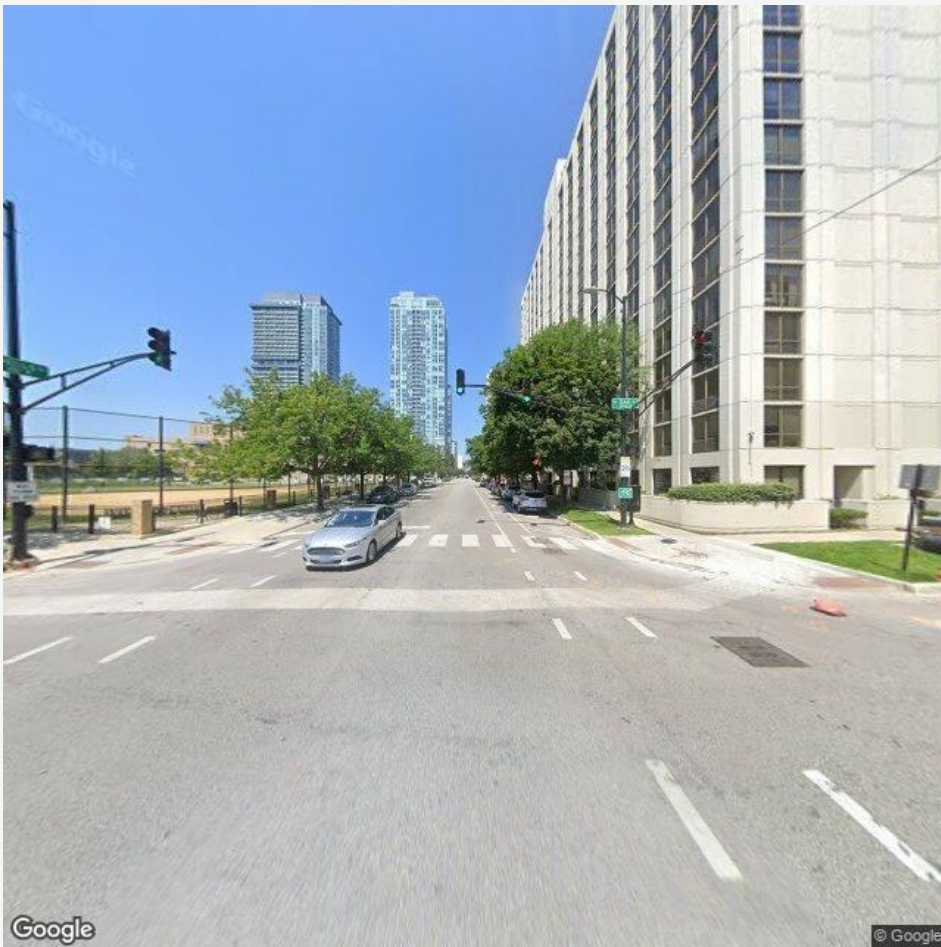
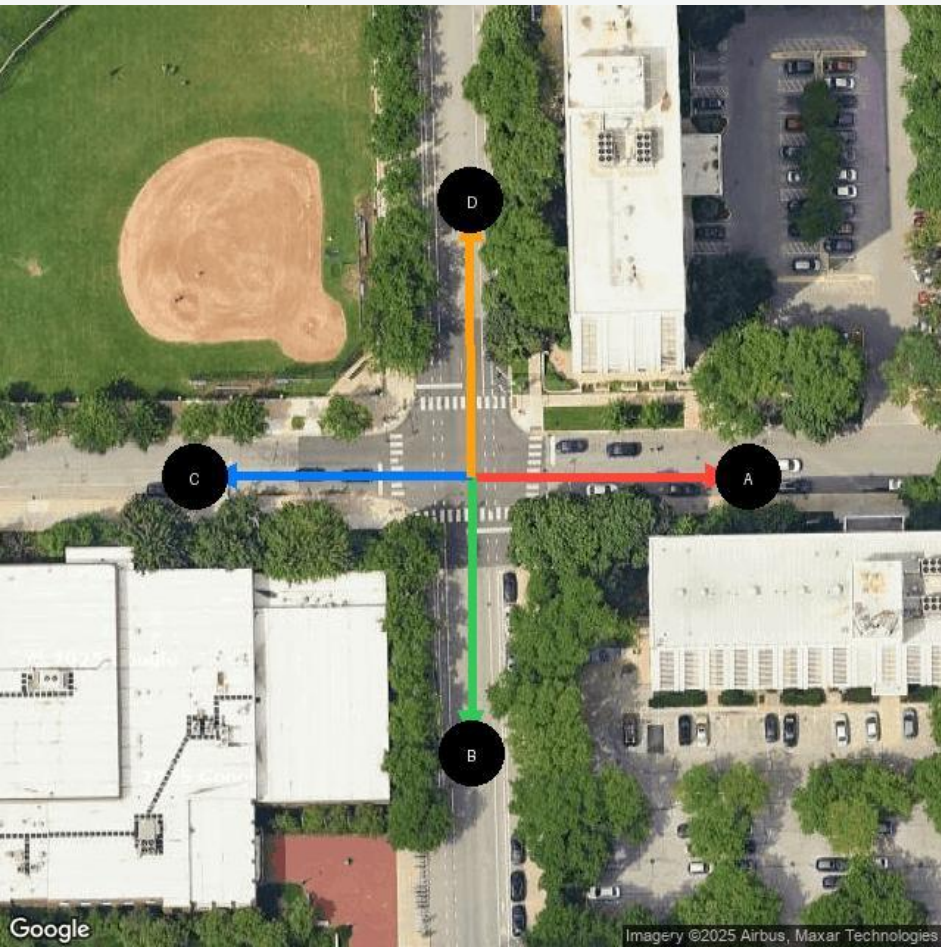


m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning

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The Task

Given a **north-up overhead map** with labeled candidate directions and a **Street View photo** taken at the same intersection, infer *which labeled direction the camera was facing*. The camera sits at the intersection center; letters point outward along each road.



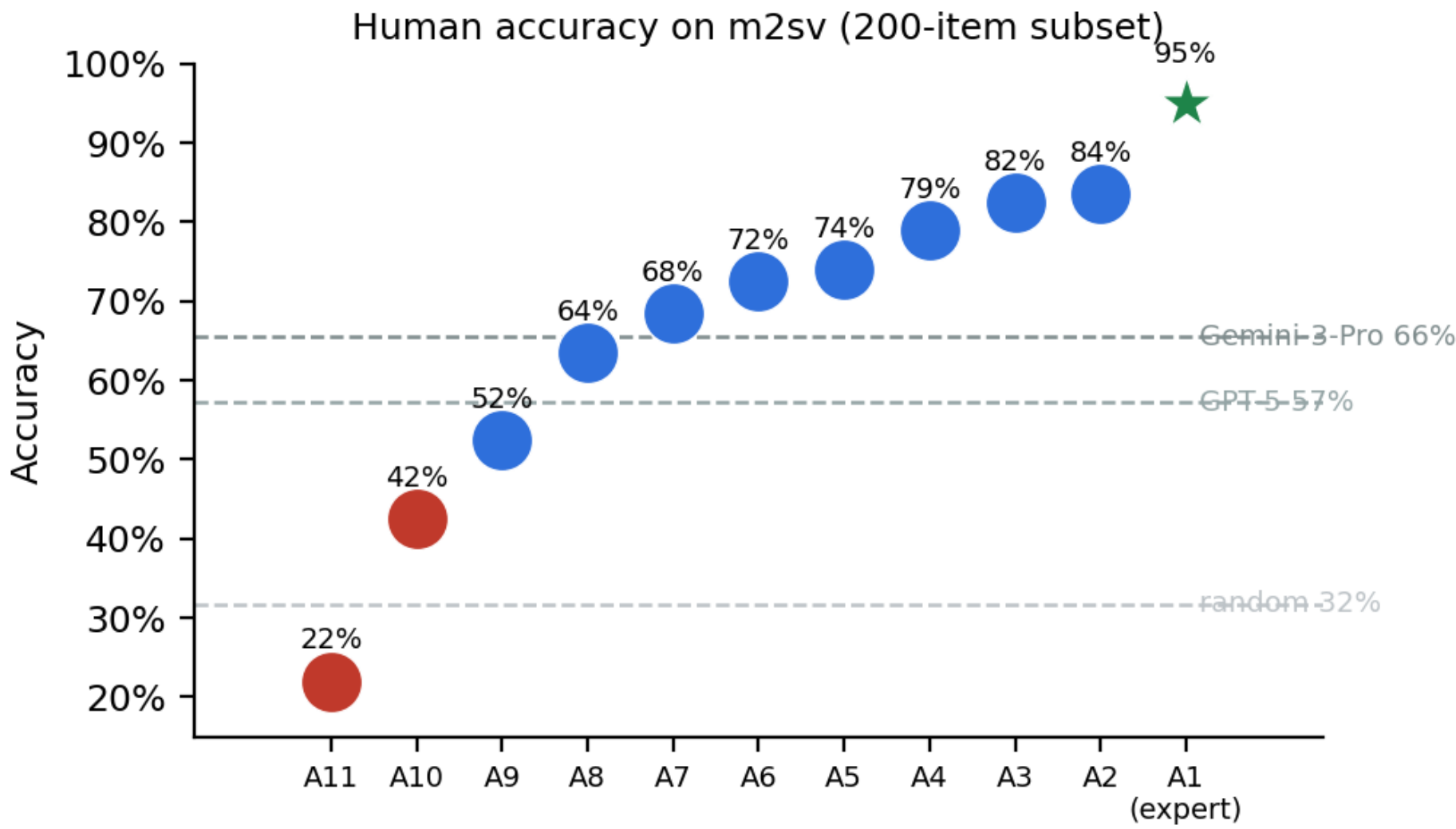
Ground truth: D (camera faces north).

Solve by aligning stable map geometry (roads, parcels, landmarks) with the ground-level view — not transient cues.

Benchmark & Data (open)

- **m2sv-20k** — 20,000 real intersections (10k train / 10k validation) across **32 cities**, multiple continents, controlled candidate ambiguity.
- **m2sv-sft-11k** — 11k curated structured reasoning traces for SFT.
- Released as **blueprints** (coordinates + azimuths) plus rendering code to respect map licensing; all eval/training code public.

Human baseline & agreement



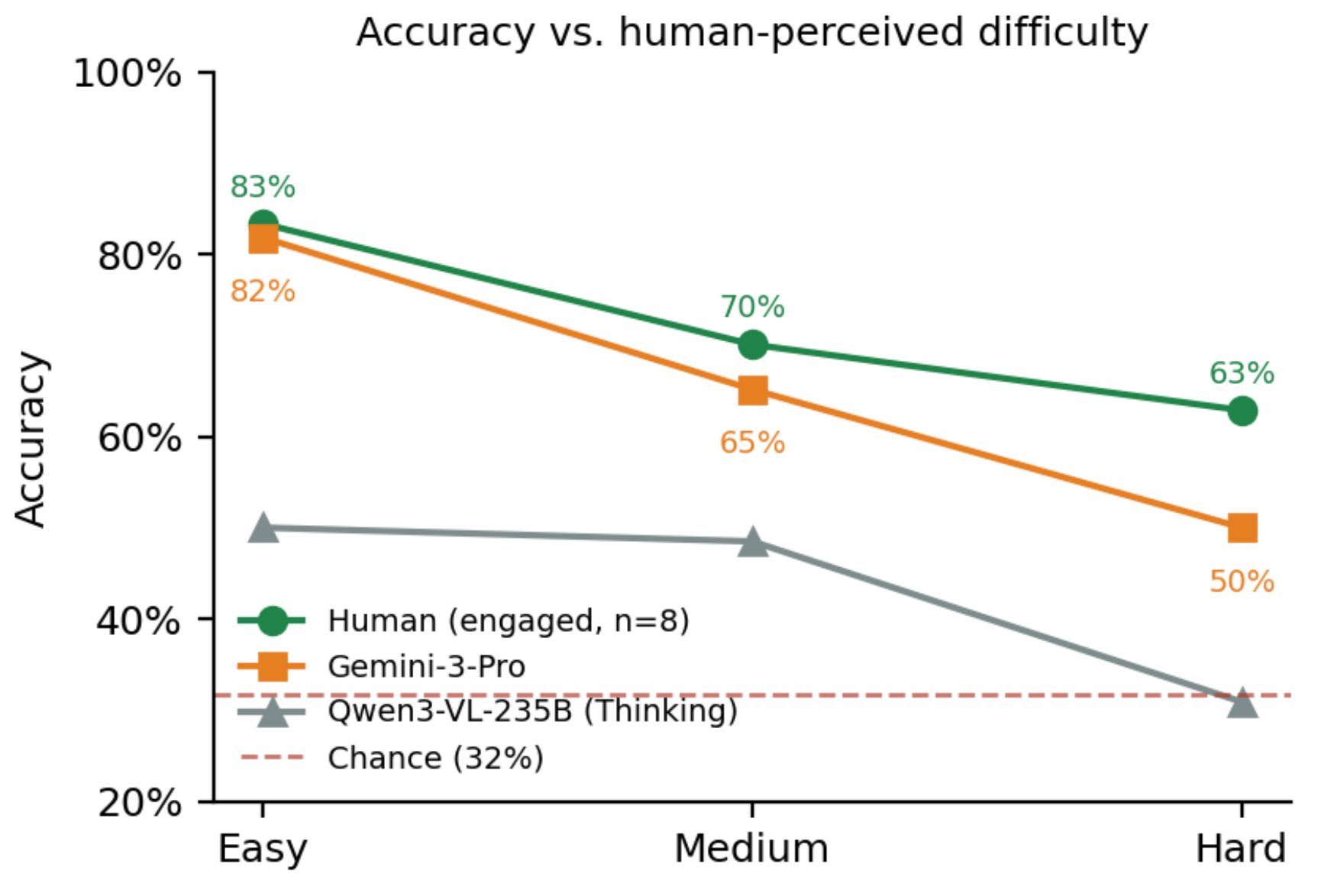
Attentive humans converge (median pairwise $\kappa = 0.48$, up to 0.76); Gemini-3-Pro sits *below* the human consensus (median $\kappa = 0.42$), Qwen3-VL-235B lower still (0.15). The gap shows up in *agreement structure*, not only accuracy.

Zero-shot: humans clear the best VLM

Model	N	Acc. (%)
Gemini-3-Pro	1k	65.2 ± 3.0
GPT-5	1k	57.2 ± 3.1
Gemini-2.5-Pro	1k	47.2 ± 3.1
Qwen3-VL-235B (Think)	1k	42.7 ± 3.1
Random baseline	1k	31.4 ± 2.9
Human (engaged, n=8)	200	72.0 ± 9.7
Human (expert)	200	95.0

Multi-annotator study (12 participated; the 10 who finished all 200 are analyzed). Two low-agreement outliers excluded by *agreement* ($\kappa_{\text{expert}} < 0.3$), not accuracy.

Difficulty: the hardest-tier collapse



Difficulty bins from **human response time**; accuracy from a model-independent axis. Humans degrade gently (83 → 63%) and stay above both models; models lose the most ground on the hardest tier, falling to (Qwen) or toward (Gemini, 50%) chance. **Symmetric** intersections (azimuth-gap ratio $S = \frac{\max}{\min}$) are hardest for all models.

Takeaways

- m2sv isolates a scalable **map↔street-view alignment** primitive frontier VLMs have *not* solved (best 65% vs human 72/95%).
- Humans beat the best model *and* agree with one another; models lie outside that consensus and collapse on the hardest, symmetric cases.
- SFT+RL on m2sv helps (+10 pts) but transfer is limited — failures are about *adapting reasoning to ambiguity*, not missing data.

Seven qualitative failure modes — every case is solved by humans, missed by the model

Errors come from *brittle alignment and reliance on unstable evidence*, not missing information. Each card: overhead map, street view, GT/prediction, and the error.

Ego↔allo inversionGT: A Pred: C

Map shows a *left* curve; the model reasons as if it curved right. Left/right flip at the map↔photo boundary.

Unstable visual cuesGT: B Pred: D

Eliminates on roof color, which is lighting- and angle-dependent and weak in overhead imagery. A human discounts it.

Landmark misbindingGT: A Pred: B

Attributes a pool to the corner house — it is two houses down. Binds a landmark to the wrong spatial anchor.

False landmarkGT: D Pred: A

Invents a nonexistent structure (underpass/bridge) and reasons as if it were present in the map.

Spatial inconsistencyGT: A Pred: C

Treats the two sides of one street as having different road widths, instead of enforcing a single coherent layout.

Internal contradictionGT: D Pred: C

Identifies left-right relations correctly, then contradicts its own assignment mid-reasoning — loses spatial state.

Symmetry trapGT: D Pred: B

Narrows to two symmetric candidates but will not escalate to fine cues (lane counts, curb geometry) to break the tie.